Buddy AI - Hardware Pin Map

Locked stack: Raspberry Pi 3 B+ (brain), ESP32-CAM (eyes over Wi-Fi), ESP32-WROOM (motors, pan-tilt servos, distance sensor), Bluetooth speaker (voice out), phone/laptop mic for v1. No Raspberry Pi Camera Module.

1. Power

From	To	Notes
Pi 5V PSU / power bank	Pi 3 B+ power input	Do not power motors from the Pi
5V supply (>=500mA)	ESP32-CAM 5V + GND	Prefer solid 5V; weak USB can brown out
Motor battery (+)	Motor driver VS / VCC motor	Match motor voltage (often 6-12V)
Motor battery (-)	Motor driver GND	
Regulated 5V	ESP32-WROOM VIN / 5V	Buck from motor pack if needed
ESP32-WROOM GND	Driver GND	Must share ground with driver
Servo 5V supply	Servo red wires	Do NOT use ESP32 3.3V for servos
Servo GND	ESP32-WROOM GND	Common ground
Ultrasonic VCC	5V	HC-SR04 prefers 5V
Ultrasonic GND	ESP32-WROOM GND	Common ground

2. ESP32-CAM (eyes) - AI-Thinker style

Camera pins are fixed on the module (firmware uses CAMERA_MODEL_AI_THINKER). You only wire power + Wi-Fi credentials. Pi fetches JPEG from http://CAM_IP/capture.

ESP32-CAM	Connect to
5V	5V supply (>=500mA)
GND	Power GND
Wi-Fi	Same network as Pi 3 B+ and ESP32-WROOM
FTDI RX/TX/GPIO0	Only when flashing firmware

Do not attach motors to the CAM board. Mount CAM on pan-tilt mast; servos are driven by ESP32-WROOM, not the CAM.

After boot, serial monitor @ 115200 prints CAM IP. Set pi-brain .env CAM_URL to http://THAT_IP (capture path is /capture, stream on port 81 /stream).

3. ESP32-WROOM -> Motor driver (firmware defaults)

Left motor = A, Right motor = B. Pins match esp32-motor/config.h.

ESP32 GPIO	L298N	L9110	Function
GPIO 26	IN1	A-IA	Left / Motor A dir 1
GPIO 27	IN2	A-IB	Left / Motor A dir 2
GPIO 25	ENA (PWM)	(unused*)	Left speed PWM
GPIO 33	IN3	B-IA	Right / Motor B dir 1
GPIO 32	IN4	B-IB	Right / Motor B dir 2
GPIO 14	ENB (PWM)	(unused*)	Right speed PWM

ESP32 GPIO	L298N	L9110	Function
GND	GND	GND	Common ground

* For L9110 / drivers without ENA/ENB: set MOTOR_HAS_PWM to 0 in config.h. Driver OUT1/OUT2 -> left motor; OUT3/OUT4 -> right motor. If a side runs backward, swap that motor's two wires or set MOTOR_INVERT_LEFT/RIGHT.

4. Pan-tilt servos -> ESP32-WROOM

Servo wire	Connect to
Brown / Black (GND)	ESP32-WROOM GND
Red (VCC)	External 5V
Orange / Yellow signal - Pan	GPIO 18
Orange / Yellow signal - Tilt	GPIO 19

5. Distance sensor -> ESP32-WROOM

HC-SR04 ultrasonic (recommended defaults):

HC-SR04	ESP32-WROOM
VCC	5V
GND	GND
TRIG	GPIO 12
ECHO	GPIO 13 (use resistor divider 5V->3.3V if possible)

If using VL53L0X ToF instead:

VL53L0X	ESP32-WROOM
VCC	3.3V
GND	GND
SDA	GPIO 21
SCL	GPIO 22

6. Audio / network (no GPIO)

Device	Connection
Bluetooth speaker	Pair in Raspberry Pi OS (voice out)
Phone / laptop mic (v1)	Wi-Fi to Pi web UI / POST /command
ReSpeaker 2-Mic HAT (next version)	Onboard voice-in - not wired in v1
Pi Camera Module	Not used - you do not have one

7. Network map

Device	Role	Example
Pi 3 B+	Brain	Runs pi-brain; opens :8080
ESP32-CAM	Eyes	http://192.168.x.y/capture
ESP32-WROOM	Motors/servos/sensor	http://192.168.x.z/cmd

All three must be on the same Wi-Fi. Put CAM_URL and MOTOR_URL in pi-brain/.env.

8. ESP32-WROOM pin summary

GPIO	Function
26	Motor A IN1
27	Motor A IN2
25	Motor A ENA (PWM)
33	Motor B IN1
32	Motor B IN2
14	Motor B ENB (PWM)
18	Pan servo signal
19	Tilt servo signal
12	Ultrasonic TRIG
13	Ultrasonic ECHO
21 / 22	ToF SDA / SCL (if VL53 instead of HC-SR04)
GND	Common with driver, servos, sensor

9. Safety notes

- Never feed motor current through the Pi 5V rail.
- Give ESP32-CAM a solid 5V supply (brownouts cause Wi-Fi drops / bad images).
- Share GND between ESP32-WROOM, motor driver logic, servos, and sensors.
- Level-shift HC-SR04 ECHO (5V) down to 3.3V for ESP32 safety when possible.
- Servos need a solid 5V supply; brownouts cause random resets.
- Put Pi 3 B+, ESP32-CAM, and ESP32-WROOM on the same Wi-Fi network.

10. Firmware / config references

Camera firmware: esp32-cam/
Motor pins: esp32-motor/config.h
Wiring notes: docs/wiring.md
Repo: https://github.com/AnuragShirodkar/Buddy-AI